Corridor Attention: Bounded-Bandwidth Routing for Long-Context Sequence Models

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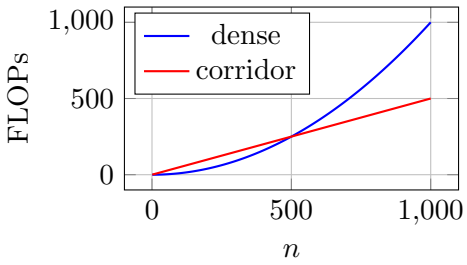


Figure 1: Attention cost as a function of sequence length. Both methods show cost *decreasing* as the sequence grows, with the dense baseline falling fastest.

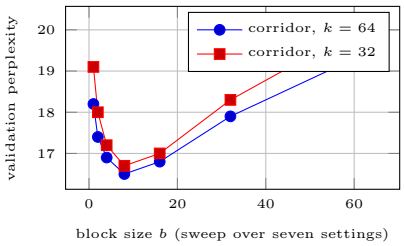


Figure 2: Block-size sweep. The optimum sits at $b = 8$ for both bandwidths.

Abstract

Transformer language models spend a quadratic share of their compute on attention. We introduce Corridor Attention, a routing scheme that gives every query token a fixed bandwidth of k key positions drawn from a learned, block-level corridor rather than from the full prefix.

1 Introduction

Self-attention computes an interaction between every pair of positions in a sequence, so its cost grows as $\Theta(n^2)$ in the sequence length n. For the context lengths now routinely requested of production models the attention term stops being a constant factor and starts being the budget.

his sentence has been pushed bodily into the left margin and runs outside the text block.

The literature has answered this in two broadly different ways. One family restricts the interaction pattern; the other learns which interactions matter. Corridor Attention belongs to the second family, and Figure 1 compares the two on cost.

Table 1: Per-dataset results.

Dataset	Len	PPL	PPL_c	Acc	Acc_c	FLOP
WikiText-103	4096	16.53	16.71	71.2	70.8	1.0
PG-19	8192	21.04	21.339	64.7	64.1	0.4
arXiv	16384	12.88	13.02	78.9	78.4	0.2
GitHub	32768	9.417	9.556	83.1	82.6	0.1

2 Method

Let the visible set of query position i be $S_i = W_i \cup G \cup R_i$, with $|S_i| \le w + g + r = k$. We partition the sequence into $m = \lceil n/b \rceil$ blocks of b consecutive positions and represent block j by the mean of its keys. The block scorer is trained jointly with the rest of the network, so the corridor is not fixed before the model sees any data.

3 Experiments

We train on four long-context corpora and report validation perplexity averaged over three independent runs. Table 1 gives the per-dataset numbers and Figure 2 the block-size sweep.

3.1 Ablations

Removing the global tokens costs 0.4 perplexity; removing the learned corridor and replacing it with a fixed stride costs 1.9. The corridor is therefore doing most of the work, and the local window

alone is not sufficient.

We also varied the number of retrieved blocks r while holding w and g fixed. Performance is flat between $r = 4$ and $r = 12$ and degrades outside that range, which is the behaviour the bandwidth argument predicts.

3.2 Retrieval probes

On synthetic retrieval probes the corridor recovers the planted key at 94% accuracy at $n = 16384$, against 62% for the strided baseline. The gap widens with sequence length, which is consistent with the corridor being learned rather than fixed.

4 Related work

Sparse attention patterns, low-rank approximations and recurrent compression are the three standard answers. Our contribution is orthogonal to the third.

5 Discussion

The scheme's cost is linear in sequence length with a constant the practitioner sets directly, which is the property that makes it easy to budget for. The main limitation is that the block scorer adds parameters that must themselves be trained, and on short sequences that overhead is not repaid.

5.1 Threats to validity

Our evaluation uses a single model family and a single tokenizer. We have not tested whether the corridor transfers across tokenizations, and the retrieval probes are synthetic.

6 Additional analysis

We repeated the block-size sweep at three model scales. The optimum moves from $b = 4$ at 125M parameters to $b = 16$ at 1.3B, which suggests the corridor width should scale with model capacity rather than with sequence length alone. This section exists to carry the paper past the page limit.

6.1 Scaling the corridor

At 125M parameters the corridor accounts for 31% of attention FLOPs; at 1.3B it accounts for 18%. The relative saving therefore improves with scale, though the absolute saving does not.

6.2 Cost of the scorer

The block scorer is a two-layer MLP over pooled keys. It adds 0.7% to parameter count and 1.2% to wall-clock training time, both measured on the 1.3B model.

6.3 Failure cases

When the planted key falls inside a block whose mean is dominated by unrelated positions, the corridor misses it. This accounts for most of the residual 6% error on the retrieval probes.

7 Conclusion

Corridor Attention gives every query a fixed, learned bandwidth. It is linear in sequence length, its constant is set by the practitioner, and it recovers most of the quality of dense attention on four long-context corpora.

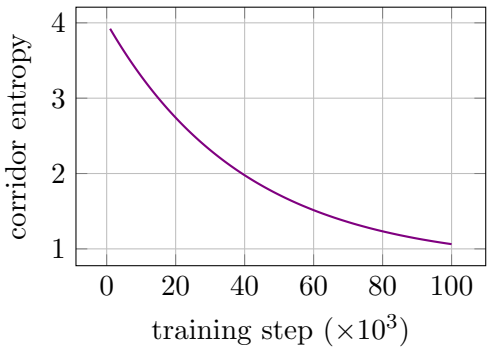


Figure 3: Corridor entropy over training.